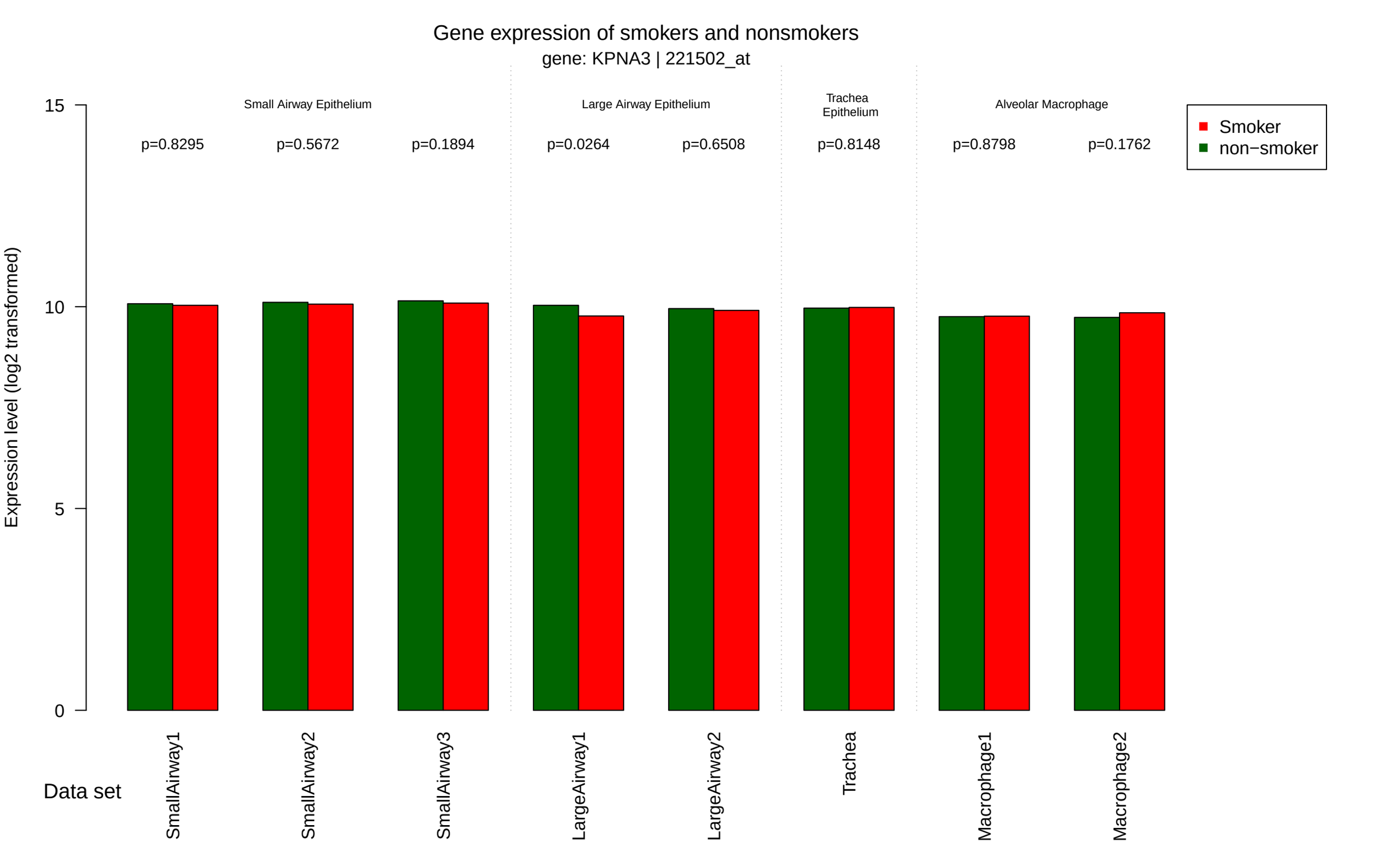
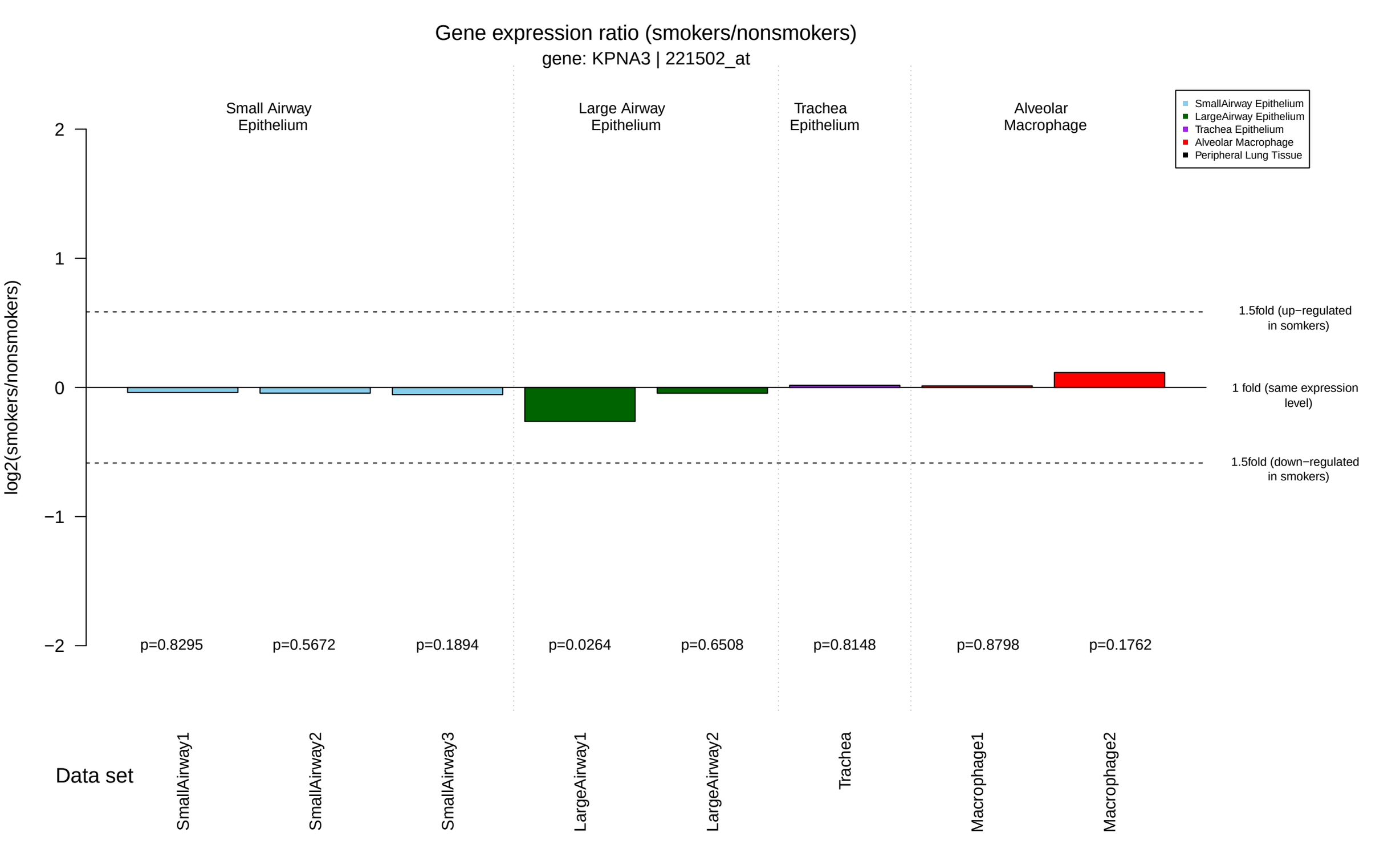
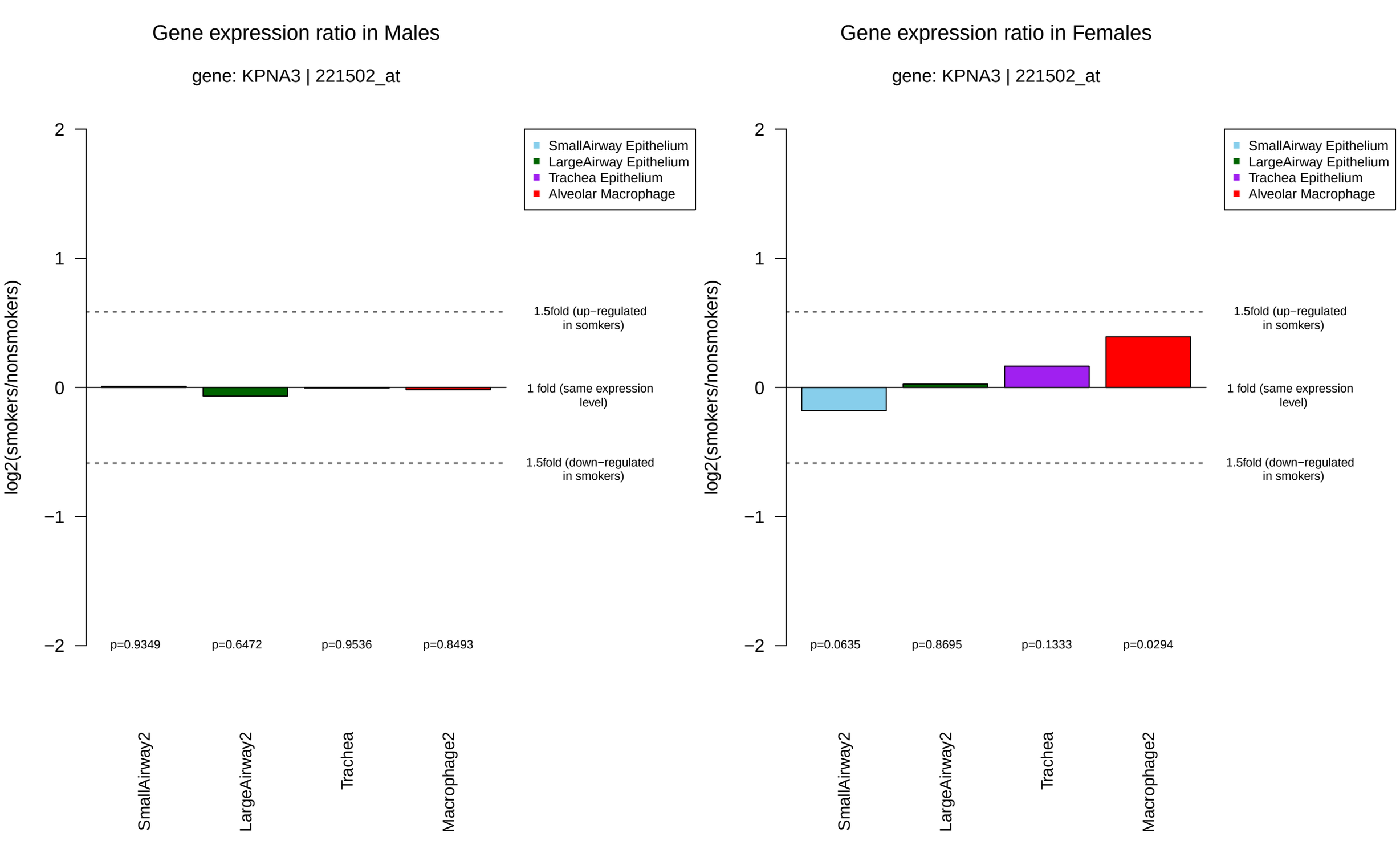
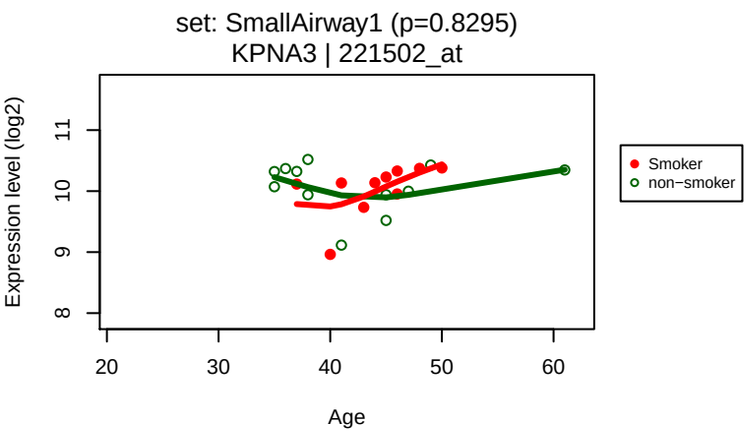
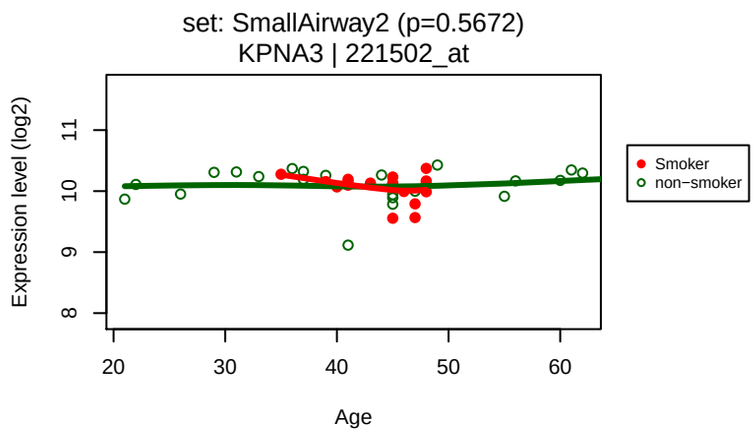
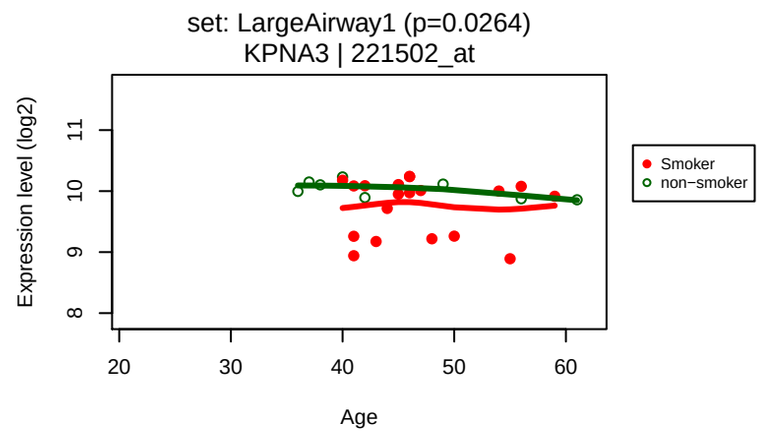

Gene expression of smokers and nonsmokers
gene: KPNA3 | 221502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8295
p=0.5672
p=0.1894
p=0.0264
p=0.6508
p=0.8148
p=0.8798
p=0.1762
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

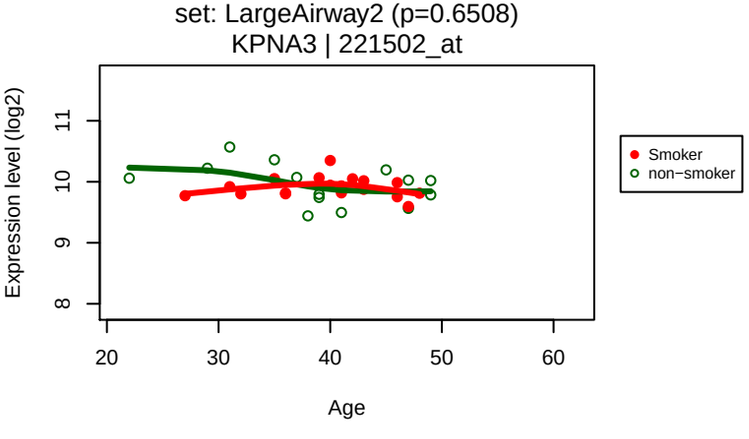

Gene expression ratio (smokers/nonsmokers)
gene: KPNA3 | 221502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8295
p=0.5672
p=0.1894
p=0.0264
p=0.6508
p=0.8148
p=0.8798
p=0.1762
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

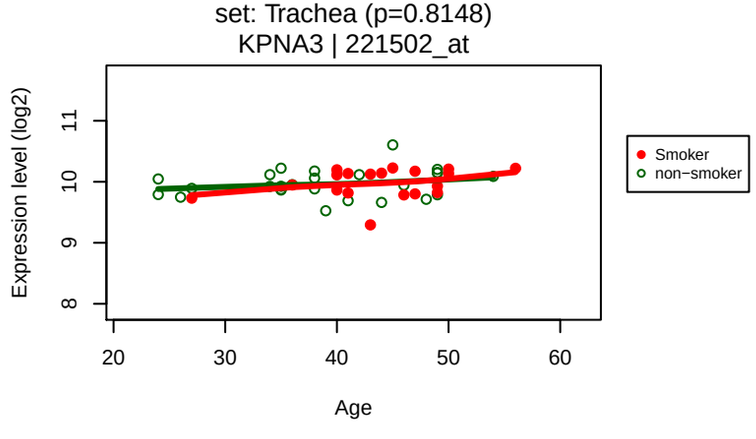

Gene expression ratio in Males
gene: KPNA3 | 221502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9349
p=0.6472
p=0.9536
p=0.8493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA3 | 221502_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0635
p=0.8695
p=0.1333
p=0.0294
SmallAirway2
LargeAirway2
Trachea
Macrophage2

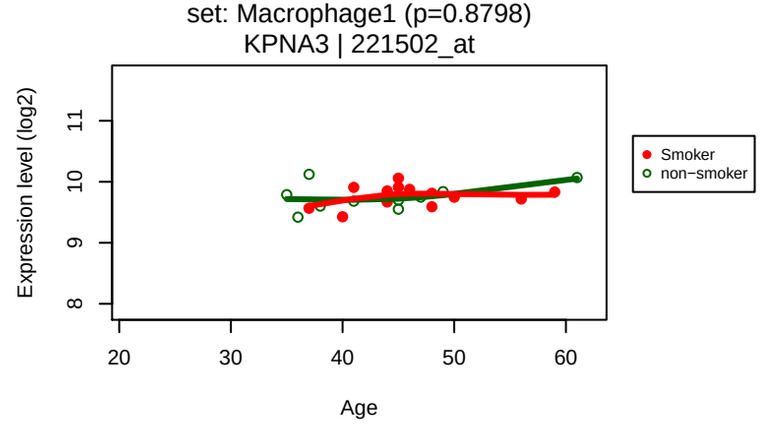

set: SmallAirway1 (p=0.8295)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker

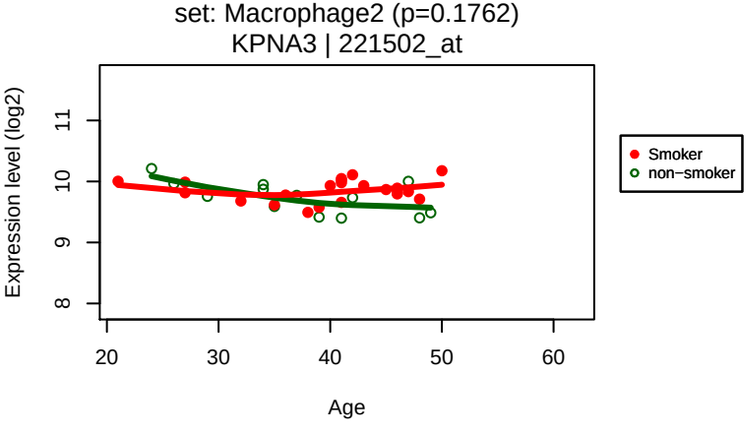

set: SmallAirway2 (p=0.5672)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker

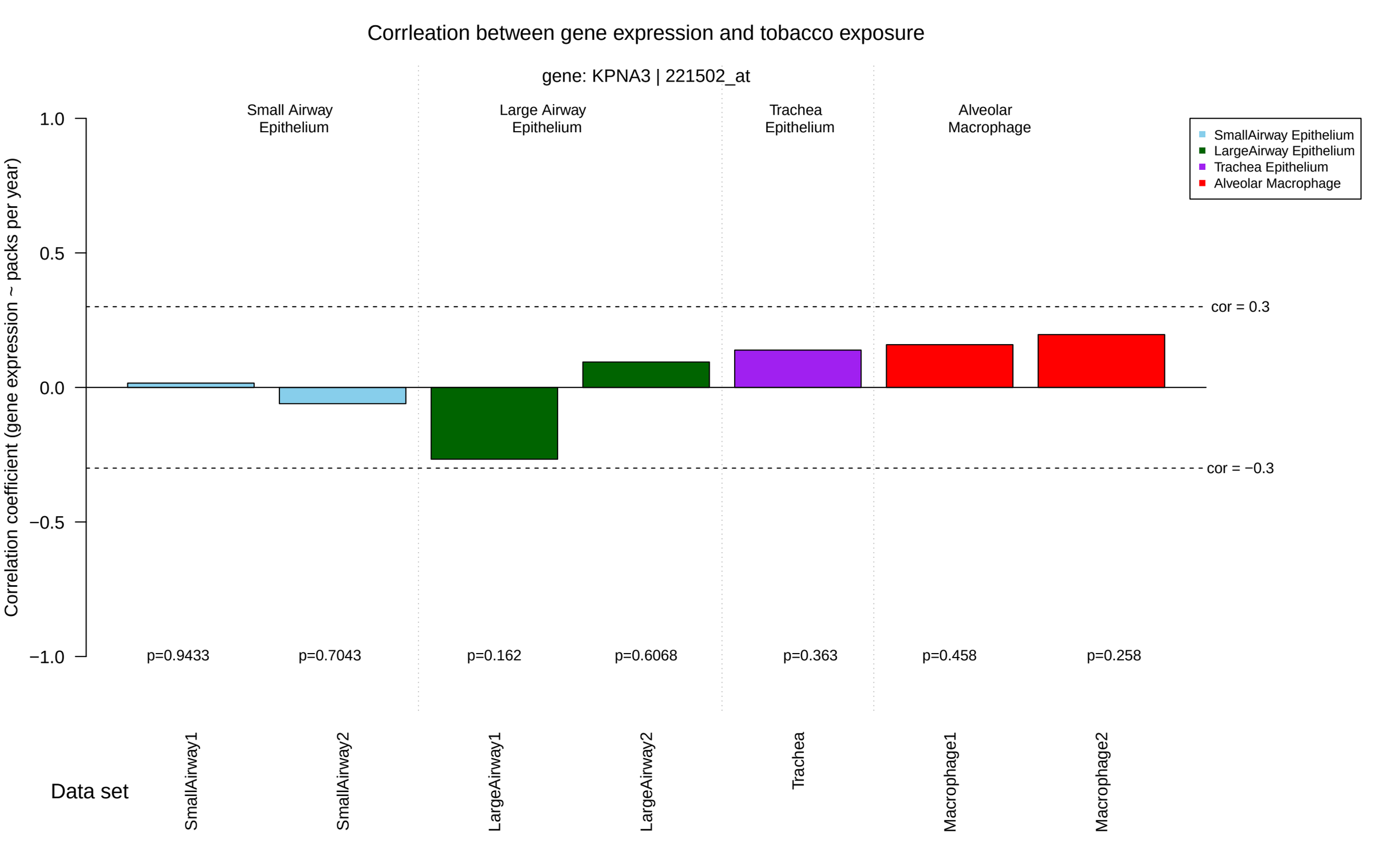

set: LargeAirway1 (p=0.0264)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6508)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8148)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8798)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1762)
KPNA3 | 221502_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA3 | 221502_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9433
p=0.7043
p=0.162
p=0.6068
p=0.363
p=0.458
p=0.258
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2